Information-Set Conditional Integrity Auditing for Hybrid Quantum-Classical Kernel Workflows

Roberto Fernández-Barrios, Iker Pastor-López, Amaia Pikatza-Huerga, and Pablo García Bringas

Abstract—Integrity auditing asks whether the evidence exposed at a workflow boundary can identify a consequential intervention. We formalize information-set conditional auditability and connect interventions, evidence regimes, sensor coverage, blind regions, and response. For a fixed deterministic predictor, evaluation- label changes leave feature and prediction statistics invariant; when class counts are preserved, label-marginal statistics are invariant as well. These boundaries support an integrated coverage methodology, not a standalone theorem. We evaluate a classical support-vector classifier and three empirically distinct fidelity-kernel profiles under 18 controlled interventions. A frozen CICIDS2017 gate and 360 exact-statevector expansion jobs cover CICIDS2017, UNSW-NB15, and ToN-IoT across scale and five fixed out-of-distribution settings. After verified deduplication, the expansion contains 11,400 model-intervention observations. All 3,600 evaluation-label observations leave features, scores, and predictions unchanged; all 1,800 prior-preserving cases also leave label marginals unchanged. Every one of the 2,184 positive conclusion impacts changes item-aligned joint-outcome evidence. A separate 165-cell simulator gate demonstrates complementary quantum-workflow coverage, including an algebraic blind region under positive-semidefinite kernel substitution. Finally, a four-contract prototype converts the coverage map into hash-chained, fail-closed `allow/hold/block` decisions. Preregistered gates show that the blind regions survive null calibration at a 5% false-alarm level and that the secondary quantum-versus-classical profile is scaler- and headroom-dependent. The contribution is the combined formalization, coverage validation, bounded quantum layer, and executable response.

Index Terms—hybrid quantum-classical systems, integrity auditing, partial observability, quantum kernels, sensor coverage

1 Introduction

Hybrid quantum-classical software distributes one reported result across heterogeneous computational and evidential boundaries. Classical records are acquired, cleaned, projected, and encoded; a circuit and execution stack produce quantum evidence; a classical learner turns a kernel or measurement into a decision; and an evaluation layer aligns that decision with reference outcomes and reports a metric. A plausible final number may therefore coexist with a failure in the data, circuit, kernel, prediction, label, or reporting path. Security analyses of hybrid systems accordingly call for lifecycle-aware controls rather than trust in an isolated algorithm [1], [2].

Robustness benchmarking usually collapses this chain into a performance change. That endpoint is necessary, but it cannot identify what changed or whether the available evidence could reveal the change. A large loss can be easy to diagnose because several independent sensors react. A smaller loss can be the more serious integrity failure when it changes the reported conclusion while every sensor available to the evaluator is invariant. Conversely, a drift statistic can react although a fixed decision is unchanged. Impact, observability, and response are distinct axes.

Consider the evaluation boundary. A deployed predictor receives the same feature batch and emits the same predictions, but the labels used to score that batch are corrupted. Balanced accuracy can change although the model has not degraded. Feature and prediction monitors cannot observe the event because their inputs are identical. Label counts expose an unbalanced corruption, but a balanced exchange of class identities preserves that marginal. Detection then requires item-aligned outcome-prediction evidence, authenticated label provenance, or an assumption that excludes the transformation. Calling the event simply stealthy omits the decisive qualifier: stealthy to which auditor, with which evidence?

This paper makes that qualifier explicit. *Information-set conditional auditability* evaluates an intervention relative to the evidence exposed at a workflow boundary and to a declared sensor family. It separates model-output changes from changes in the evidence used to evaluate that output, then records the residual blind region. Quantum kernels form a bounded but demanding instantiation: their workflow crosses classical preprocessing, parameterized circuit construction, fidelity estimation, a kernel matrix, a classical support-vector decision, and outcome-based evaluation. This boundary is quantum-specific without requiring a physical processor.

Three research questions organize the study. **RQ1** asks which controlled interventions are observable from feature, prediction, label-marginal, item-aligned outcome, and quantum-workflow evidence, and where exact blind regions remain. **RQ2** asks whether those coverage boundaries recur across datasets, fixed in- and out-of-distribution (ID/OOD) environments, projected dimensions, and kernel profiles. **RQ3** asks whether provenance, semantic, algebraic, repeated-estimation, and output sensors provide complementary

- *The authors are with the Faculty of Engineering, University of Deusto, Avda. de las Universidades 24, 48007 Bilbao, Spain. E-mail: roberto.fernandez.b@deusto.es (corresponding author), iker.pastor@deusto.es, a.pikatza@deusto.es, pablo.garcia.bringas@deusto.es.*
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quantum-boundary coverage and whether their evidence can drive a fail-closed decision.

The contribution is one integrated package:

- a workflow formalization that separates model-output, evaluation- conclusion, and execution-integrity effects and conditions every auditability claim on an information set;
- exact blind-region statements and an attack-to-sensor coverage method that separates structural invariance from low empirical detector power;
- multi-dataset and fixed-OOD validation with verified computational-unit deduplication and uncertainty clustered at the data-split level;
- a bounded quantum-specific layer combining circuit/kernel provenance, semantic comparison, algebraic validation, repeated estimation, and output monitoring; and
- an executable four-contract prototype whose `allow/hold/block` policy makes the coverage map operational within a local research boundary.

No constituent is claimed to be individually sufficient novelty. The two propositions make the evidence boundary exact; contracts, hash chains, and semantics-preserving tests have clear prior art; and the observed ZZ-versus-RBF profile is neither a quantum-advantage result nor a universal vulnerability ordering. Physical hardware, calibrated backend noise, scheduling, provider-side security, multi-tenant interference, and operational Fleet Management are outside the claim.

2 Related Work and Positioning

2.1 Hybrid Integrity and Quantum Software Assurance

Volya *et al.* decompose classical-quantum systems into classical control, compilation, quantum execution, and measurement surfaces [1]; surveys extend the threat space to malicious compilation, hardware faults, cloud exposure, side channels, and classical dependencies [2], [3]. Quantum Vulnerability Factor quantifies circuit-output sensitivity to injected faults [4], while Quantum Leak demonstrates timing leakage in cloud services [5]. These works establish impact and threat surfaces, but do not condition a workflow-audit claim on the exact evidence exposed to its sensor.

The closest direct overlap is QCIVET, a concurrent 2026 preprint that combines stage contracts, hash-chained traces, behavioral subtyping, semantic observable checks, calibrated-noise models, and real-QPU validation [6]. We therefore make no priority claim for contracts, hash chains, stage decomposition, or fail-closed verification. Our unit of analysis is instead the information set that makes a changed asset distinguishable: evaluation labels and conclusions are first-class boundaries, coverage is mapped across multiple data environments, and the contract merely executes that map. A second concurrent line evaluates structural, interaction, and behavioral circuit-integrity scores [7]; it likewise establishes metric complementarity, but not the cross-boundary evidence lattice studied here. QML-PipeGuard, a concurrent preprint by one of the QCIVET authors, fingerprints the quantum stage of a QML pipeline through expectation values under a tomographically structured measurement family, absorbs benign calibration drift within a calibrated tolerance, and detects channel substitution on an IBM Heron processor [8]. Its unit is the quantum channel: it does not model the evaluation-label path, provenance, or coverage across data environments, and its tolerance is a hardware-drift envelope rather than the information-set conditioning studied here.

Quantum software testing already addresses the oracle problem through differential or semantics-preserving variants. QDiff, MorphQ, and QEMI test quantum stacks using cross-implementation, metamorphic, or equivalence-modulo-input relations [9]–[11]. Compiler verification, program logics, runtime assertions, and design-by- contract provide complementary formal and runtime mechanisms [12]–[16]. Our benign transpilation and common-unitary conditions are therefore negative controls, not a new compiler-testing method. Their role is to expose what a provenance hash, semantic probe, algebraic check, repeated estimate, or final prediction can actually establish about one hybrid kernel workflow.

2.2 Monitoring, Labels, and Benchmark Validity

Classical monitoring work makes clear that labels change what can be inferred. Black-box shift estimation targets label shift under source-target assumptions [17]; MLDemon and uncertainty-aware monitoring combine unlabeled observations with selectively acquired labels [18], [19]. A concurrent evidence-sufficiency framework maps proxy coverage under delayed ground truth and formalizes the inability of feature proxies to expose pure concept drift when the feature distribution is unchanged [20]. Independently, Bajaj names *evaluation blindness*, a measurement that is indistinguishable from a healthy state while the system fails, and documents its prevalence in a corpus of production incidents [21]. That taxonomy motivates our question; it does not formalize information regimes, derive exact blind regions, validate sensor coverage, or compose a response.

Our label intervention is different: it changes the evaluation labels attached to fixed items while holding input and predictor fixed. Because the label source may itself be compromised, label availability is not sufficient; the audit must separate an unauthenticated marginal from trusted item-aligned outcomes. Likewise, MMD, KS, and Jensen–Shannon statistics are useful signals [22], but a nonzero value is not automatically a calibrated alarm. Thresholds, false-alarm control, and power are deployment-specific.

Security-oriented machine-learning benchmarks are vulnerable to sampling, labeling, metric, baseline, and threat-model errors [23]. Quantum- kernel benchmarks add feature-map, encoding, preprocessing, and comparator dependence [24]. Work on adversarial quantum kernels and QML poisoning asks whether attacks change performance and whether defenses recover it [25], [26]. Our ZZ-versus-RBF comparison is retained only to show that impact profiles vary by frozen pipeline.

Table 1 states the combination claim. We found no previous study that jointly conditions integrity auditability on a partially ordered information set, models the evaluation-label path explicitly, separates exact and empirical sensor blind regions, validates their transport in a quantum- kernel workflow, and drives a fail-closed response. Removing the conditioning would reduce the study to a detector

TABLE 1
Positioning against the closest lines of work. A dash means that the property is not the work's declared object, not that the work is deficient.

Line of work	Evidence conditioning	Evaluation-label boundary	Quantum evidence	Validation and response
Hybrid threat taxonomies [1], [2]	–	–	Broad lifecycle	Taxonomy; no executable response
QCIVET [6]	Stage contracts/observables	–	Calibrated noise and QPU	Three applications; verification engine
MorphQ/QDiff/QEMI [9]–[11]	Semantic relations	–	Software stacks	Cross-platform test verdicts
Evidence sufficiency [20]	Proxy-coverage dimensions	Delayed outcomes	–	One dataset; readiness gate
Circuit-integrity metrics [7]	Fixed metric layers	–	Circuit metrics	Circuit corpus; metric complementarity
QML-PipeGuard [8]	Calibrated observable tolerance	–	QSVM channel on IBM Heron	Two-qubit QSVM; drift versus substitution
Evaluation blindness [21]	Detectability taxonomy	Silent measurement failures	–	50 incidents; failure budget
This work	**Explicit evidence partial order**	**First-class protected boundary**	**Bounded simulator gate**	**Three datasets, eight environments, local fail-closed response**

benchmark; claiming the contract or the propositions alone would overlap established work.

2.3 Relation to Companion Work

Three companion studies by the authors share infrastructure or vocabulary with this article and are disclosed to delimit the claim. *Sharp target-domain certificates* [27] bounds the predictive advantage of a fixed quantum-kernel candidate over a prespecified classical family under distribution shift with partial target labels; it shares the UNSW-NB15 and ToN-IoT sources, fidelity kernels, and a fail-closed prediction lock, but its object is an identification interval for relative advantage, not the observability of interventions. *Conditional validity of quantum event classifiers* [28] is the closest conceptual relative: it also conditions fail-closed verdicts on a declared information set and exhibits exact sensor blindness, there to weight-only nuisances in collider data under systematic and finite-shot uncertainty. The present article applies the conditioning to a different object, the integrity of the evidence chain under controlled interventions at workflow boundaries, with provenance, coverage and executable response, rather than the certification of a scientific claim under benign systematics. *Candidate comparability before promotion* [29] asks whether a challenger should replace a deployed intrusion detector after a drift alarm; it shares the three benchmarks and the label-free monitors, but its unit is a sequential promotion decision that begins where this article ends. None of the propositions, interventions, tables or experiments reported here appears in those works.

3 Formal Model and Threat Boundary

3.1 Workflow and Impact Loci

Represent a hybrid evaluation workflow as

$$\mathcal{W} = (X, P, C, K, E, f, y, R), \tag{1}$$

where X is the acquired feature batch, P classical preprocessing, C the parameterized circuit or feature map, K the estimated kernel, E execution metadata, f the resulting predictor, y reference outcomes, and R the reported evaluation result. An intervention a may act on one or more elements of $\mathcal{W}$.

We distinguish three effects. *Model-output impact* measures changed scores or predictions on aligned items; *benchmark-conclusion impact* measures a changed reported metric or comparison; and *execution-integrity impact* measures changed circuit, kernel, or execution evidence even if the decision is invariant. The primary conclusion endpoint is

$$\Delta_R(a) = \mathrm{BA}_0 - \mathrm{BA}_a, \tag{2}$$

where BA denotes balanced accuracy. For aligned deterministic predictions,

$$\Delta_f(a) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[f_a(x_i) \neq f_0(x_i)]\,. \tag{3}$$

When $\Delta_R(a) > 0$ but $\Delta_f(a) = 0$, the measured loss lies in the evaluation or label path, not in predictor output.

3.2 Information Regimes and Exact Boundaries

The auditor receives a partially ordered family of evidence sets:

- $\mathcal{I}_X$: features or preprocessed features;
- $\mathcal{I}_{XF}$: features plus scores or predictions, with $\mathcal{I}_X \subset \mathcal{I}_{XF}$;
- $\mathcal{I}_{Y_m}$: the evaluation-label marginal or class counts;
- $\mathcal{I}_{XFY}$: item-aligned features, predictions, and outcomes, containing both $\mathcal{I}_{XF}$ and the outcome marginal; and
- $\mathcal{I}_Q$: circuit, transpilation, kernel-estimation, and execution evidence.

$\mathcal{I}_{Y_m}$ is not an intermediate step between $\mathcal{I}_{XF}$ and $\mathcal{I}_{XFY}$, and $\mathcal{I}_Q$ is a separate branch that may be joined with classical evidence. A sensor S is admissible in regime $\mathcal{I}$ only if it is a measurable function of evidence in $\mathcal{I}$. Auditability is therefore reported as $\mathsf{A}(a \mid \mathcal{I}, S)$, not as an unqualified property of a.

Proposition 1 (feature/prediction blindness). Let f be fixed and deterministic. If T_y changes only evaluation labels, so $X' = X$ and $f'(X') = f(X)$, then every audit statistic measurable with respect to $\mathcal{I}_X$ or $\mathcal{I}_{XF}$ is invariant under T_y.

Proof: The complete arguments supplied to the statistic are identical before and after intervention. A deterministic function of identical arguments returns the same value. □

Proposition 2 (label-marginal blindness). If T_y also preserves the empirical class histogram, every statistic measurable only with respect to $\mathcal{I}_{Y_m}$ is invariant.

Proof: The statistic's sufficient input, the empirical label-count vector, is unchanged. □

Thus, a prior-preserving label-only intervention requires joint label- prediction evidence, item-level provenance, an independently authenticated label source, or an assumption restricting admissible transformations. These are finite-batch capability statements, not claims about attack prevalence.

For information regime $\mathcal{I}$, sensor family $\mathcal{S}$, and materiality threshold τ, define the auditability gap

$$\mathcal{G}_{\mathcal{I},\mathcal{S}}(\tau) = \{a : |\Delta_R(a)| \geq \tau \ \wedge\ \mathcal{S}(a) = \mathcal{S}(0)\}\,. \tag{4}$$

The absolute endpoint includes apparent improvement caused by an integrity failure; positive-impact counts below use the signed drop for interpretability. Exact equality is asserted only when the construction supplies it. For a non-invariant signal, practical detection requires a separately calibrated alarm and false-alarm budget.

3.3 Threat and Trust Boundary

This study is an integrity stress test. Its manipulations are controlled interventions, not estimates of attack frequency. Data-to-evaluation gates trust the training routine and fitted model artifact while changing evaluation features, post-projection values, or labels. The quantum gate additionally permits circuit/parameter mutation and post-estimation kernel corruption. The protected assets are the evaluation representation, circuit and parameters, kernel matrix, predictions, outcomes, and reported conclusion.

The adversary does not compromise every evidence source simultaneously. Each claim instead states what can change and what the auditor sees. This is essential: if an adversary can replace both an asset and every trusted reference used to authenticate it, the present local contract cannot identify the event. Malicious scheduling, device-calibrated noise, provider-side manipulation, multi-tenant interference, side channels, and physical-device execution are excluded.

4 METHODOLOGY

4.1 Data, Pipelines, and Interventions

The frozen first gate uses a balanced binary subset derived from CICIDS2017 [30]: 3,000 rows, 77 numeric columns before projection, and 128/128 capped training/evaluation samples. The expansion adds a 256/256 CICIDS scale gate; balanced ID subsets of UNSW-NB15 [31] and ToN-IoT [32]; four temporal CICIDS source-target pairs; and one UNSW temporal pair. Expansion jobs otherwise use 128/128 samples. Every data-to-evaluation gate evaluates projected dimensions 8, 10, and 12. The eight fixed expansion environments are enumerated in the supplement. They are not a probability sample of deployments. Balanced staging, numeric-only fields, and near-chance clean performance in several temporal pairs constrain external interpretation.

Non-finite numeric values are replaced during staging. Train-defined preprocessing is applied without evaluation labels. Each branch fits TruncatedSVD on training features. The RBF support-vector classifier (SVC) then uses training-fitted standardization; fidelity-kernel branches use frozen training-fitted angular scaling to $[0, 2\pi]$. No hyperparameter search is performed, so SVC/QSVC comparisons concern complete frozen pipelines rather than an isolated causal effect of kernel geometry.

Fidelity kernels use one feature-map repetition. Four configured labels reduce to three empirical profiles because Z and PauliXZ yield identical matrices in the frozen implementation: ZZ, PauliXYZ, and Z/PauliXZ-equivalent. Gate 1 uses Qiskit Machine Learning's ideal reference evaluator [33]. Expansion gates use a tagged exact-statevector evaluator that prepares each unique state once and computes $K(x, y) = |\langle\phi(x)|\phi(y)\rangle|^2$. Against the reference, unit matrices agree at 10^{-10}; a 128/128 replay preserves every discrete endpoint and sensor value, with maximum difference 1.222×10^{-4} confined to ROC-AUC half-tie behavior. This is an ideal-simulation optimization, not shot-noise or QPU evidence.

The intervention suite has six mechanisms at three frozen severities: feature sign flip, feature-wise mean shift, scaling drift, feature dropout, random evaluation-label flip, and prior-preserving evaluation-label flip. The first four may change model input and output. The last two keep features, scores, and predictions fixed; the prior-preserving transformation also retains the class histogram. Equal mechanism weights define an engineering stress profile, not a threat distribution.

Feature-only sensors comprise feature Jensen–Shannon divergence (JSD), maximum mean discrepancy (MMD), and the Kolmogorov–Smirnov rejection rate. Prediction- aware evidence adds score-distribution JSD. Label-marginal evidence uses prior shift and label JSD. Joint-outcome evidence includes prediction-rate, disagreement, prediction-distribution, and confusion-profile changes. Raw invariance is primary for structural-blindness claims; min–max normalization is retained only as a fixed-design sensitivity.

4.2 Experimental Units and Uncertainty

Gate 1 crosses five split seeds with four nested model/perturbation seeds. Its 7,980 raw rows include 3,420 exact repeated SVC rows emitted under different quantum-map configurations. Equality across performance, impact, and sensor columns is verified before deduplication, leaving 4,560 unique model- intervention observations.

The eight expansion environments cross the same five split seeds with two nested model seeds. All 360 expected CSV/JSON job pairs completed. Their 13,680 raw rows contain 2,280 verified repeated SVC rows, leaving 11,400 unique observations. No key is collapsed unless all relevant numeric evidence agrees.

Primary coverage endpoints are exact raw invariance/response counts in the prespecified design. The secondary model-profile endpoint is the within-split difference in suite-average balanced-accuracy conclusion impact. Nested model- seed values are averaged within each split;

the five split seeds are the inferential units. Two-sided 95% Student-t intervals use four degrees of freedom. They describe within-environment resampling uncertainty, not a deployment population, and are not multiplicity-adjusted. Analyses over all 20 split-model combinations are sensitivity checks, not 20 independent replications.

4.3 Quantum-Specific Integrity Gate

A separate simulator gate isolates the quantum boundary using CICIDS, 64/64 samples, five split seeds, projected dimensions 4, 6, and 8, and 11 conditions: 165 split-dimension-condition cells. Negative controls are clean execution, benign level-1 transpilation, and a common post-feature-map X rewrite that preserves fidelity. Mutations include data-dependent RZ gates at strengths 0.02 and 0.10, increasing feature-map repetitions, asymmetric and diagonal-eroded kernel edits, a symmetric positive-semidefinite (PSD)-preserving substitution, and binomial fidelity-estimation emulators at 256 and 1,024 shots.

Each cell records canonical OpenQASM 3 circuit and parameter provenance, kernel hashes, semantic kernel differences against a trusted reference, symmetry/diagonal/eigenvalue checks, repeated-estimation discrepancy, and prediction changes. The shot conditions sample exact fidelities: they exercise the contract under controlled stochastic estimation but are neither sampler- backend nor QPU runs. PSD repair is containment, not proof of integrity.

4.4 Executable Response Prototype

A local hybrid-software-as-a-service prototype instantiates four ordered contracts: input/preprocessing, circuit/kernel, execution/result, and evaluation/report. Each records evidence, violations, and review conditions in an SHA-256 chain. Four passes yield `allow`; a review yields fail-closed `hold`; a violation yields `block`. The chain exposes later modification but provides neither provider-authenticated identity nor non-repudiation.

The demonstrator uses one compact CICIDS cell (split 42, dimension 4, 32/32 samples) and six prespecified scenarios. It evaluates composition and decision logic, not predictive generalization or throughput.

5 Results

5.1 Exact Label-Path Blind Regions

All random and prior-preserving evaluation-label cells have zero prediction disagreement: the model sees the same X and emits the same predictions. Feature JSD, MMD, KS rejection, and score JSD are exactly unchanged. Balanced- accuracy conclusion impact is nonnegative in the executed design but not positive in every cell. Gate 1 contains 1,276 positive and 164 zero impacts among 1,440 label observations; the expansion contains 2,184 positive and 1,416 zero impacts among 3,600. No negative impact occurs.

Prior-preserving flips also produce zero prior shift and label JSD, occupying the exact blind region of Propositions 1–2. In Gate 1 at flip rate 0.10, their mean conclusion impacts range from 0.0492 to 0.0633 across evaluated model-dimension cells. Joint confusion evidence reacts whenever the conclusion impact is positive. Table 2 makes the denominators explicit.

TABLE 2
Verified coverage counts after computational-unit deduplication. $\mathcal{I}_{Y_m}$ applies only to prior-preserving rows; joint response is evaluated only among positive impacts.

Verified quantity	Gate 1	Expansion
Unique observations	4,560	11,400
Label-only / $\mathcal{I}_{XF}$ blind	1,440 / 1,440	3,600 / 3,600
Prior-preserving / $\mathcal{I}_{Y_m}$ blind	720 / 720	1,800 / 1,800
Positive impact / joint response	1,276 / 1,276	2,184 / 2,184

The expansion reproduces the information boundary without exception across all eight fixed environments and evaluated model profiles. Figure 1 shows response fractions by information regime. Random label flips provide a negative control: they remain invisible to $\mathcal{I}_{XF}$ but are visible to label-marginal monitoring. The intervention has not become intrinsically less stealthy; the auditor has acquired a more informative view.

5.2 Secondary Model-Impact Profile

QSVC-ZZ has the largest mean conclusion impact in the frozen ID suite. Relative to SVC-RBF, mean ratios are 2.524, 3.094, and 3.323 at dimensions 8, 10, and 12; paired differences, rather than ratios, are the inferential endpoint. The within-split differences are respectively 0.02427 (95% CI 0.01200–0.03654), 0.03289 (0.01998–0.04580), and 0.03855 (0.02773–0.04937), with all five split clusters positive at every dimension. Corresponding ZZ/SVC impacts are 0.04019/0.01592, 0.04860/0.01571, and 0.05514/0.01659. These intervals describe one fixed environment.

Across the eight expansion environments, mean ZZ-minus-SVC impact is positive in 7/8 environments at dimension 8 and 8/8 at dimensions 10 and 12. Environment-specific intervals exclude zero in 4/8, 7/8, and 5/8 environments, respectively. Means range from -0.00009 to 0.09382 at dimension 8, 0.00107 to 0.09508 at dimension 10, and 0.00403 to 0.11585 at dimension 12 (Fig. 2). No pooled population effect is claimed. Clean balanced accuracy and impact correlate positively across expansion cells ($r = 0.714$, exploratory), revealing performance-headroom confounding. The conditional auditability result does not depend on this model ordering.

5.3 Quantum-Workflow Coverage

All 165 cells are present and all nine prespecified acceptance checks pass at the frozen numerical tolerance (10^{-8} for nonzero response unless a stricter contract check is stated). Clean cells trigger no sensor. Benign transpilation and the common-unitary rewrite change serialized provenance while preserving semantic kernels, algebraic properties, and outputs in every cell. Thus, raw hash difference requires policy-aware adjudication; it is not evidence of harmful semantic change.

Every data-dependent circuit mutation changes the semantic kernel. The mild RZ mutation changes no evaluated prediction, the stronger mutation changes predictions in 2/15 cells, and increasing feature-map repetitions does so in 11/15.

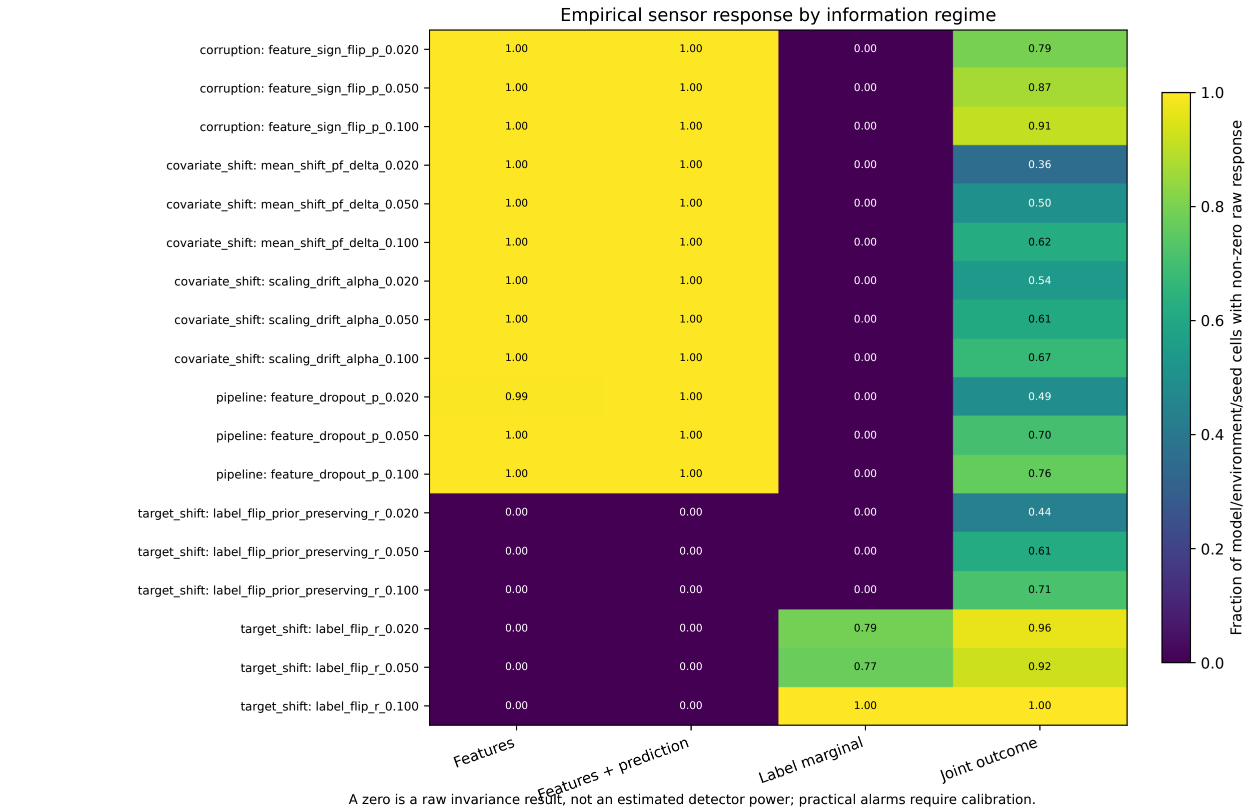


Fig. 1. Empirical sensor response by information regime. Fractions are over model/environment/seed cells. Zero denotes raw invariance, not estimated detector power; operational alarms require separate calibration.

Every asymmetric and diagonal-eroded matrix is detected algebraically but changes no prediction at the tested severity. Conversely, the PSD- preserving mixture remains symmetric, unit-diagonal, and PSD in all cells: algebraic validation is exactly blind. Trusted provenance and semantic comparison expose all 15 substitutions; output monitoring reacts in only 1/15.

Both shot emulators produce nonzero repeated-estimation discrepancy in all 15 cells. Output changes occur in 7/15 cells at 256 shots and 3/15 at 1,024 shots, with mean balanced-accuracy drops 0.00521 and 0.00313. These results support repeated-estimation logic, not a hardware or

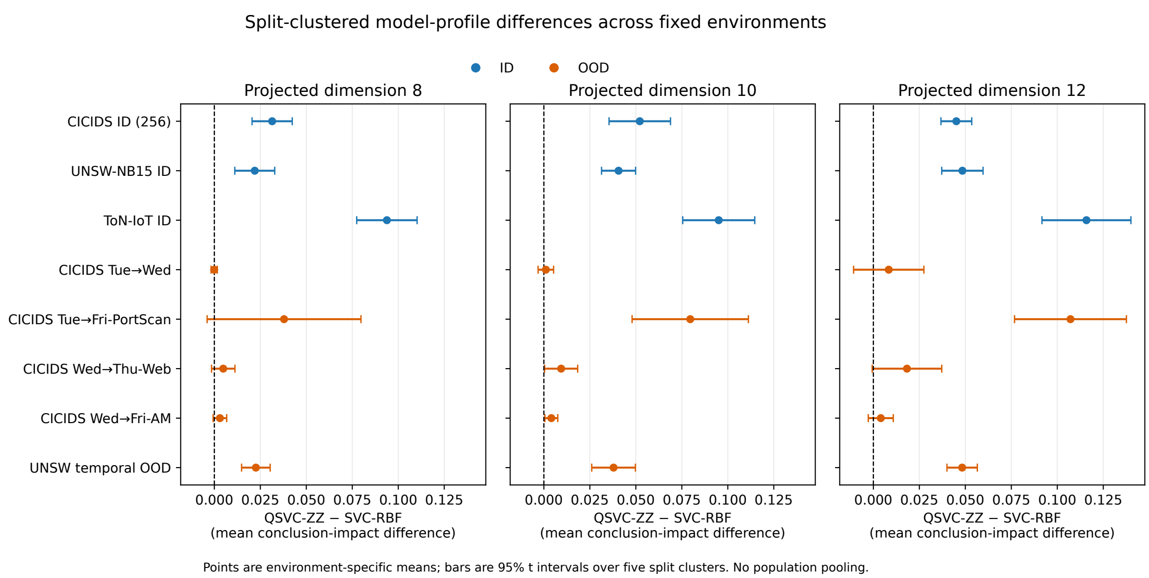
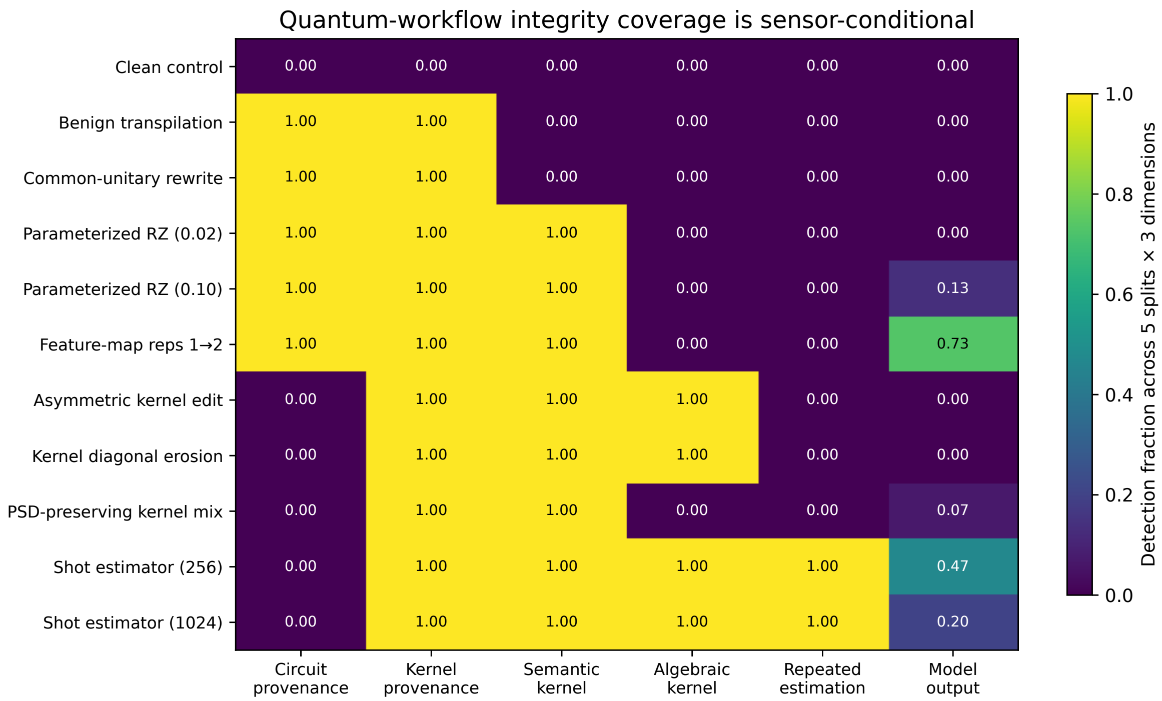


Fig. 2. Environment-specific ZZ-minus-SVC mean conclusion-impact differences and 95% t intervals over five split clusters. ID and OOD settings are fixed cases; no population pooling is performed.



Provenance assumes a trusted reference; semantic and algebraic checks answer different questions. Shot rows are binomial emulation, not QPU runs.

Fig. 3. Sensor response across clean and semantics-preserving controls, circuit mutations, kernel corruptions, and finite-shot emulators. Coverage is conditional on the evidence exposed to each sensor; output invariance does not imply workflow integrity.

calibrated-noise claim. Figure 3 summarizes the complementary sensor profile: hashes overreact to approved equivalence, algebraic checks miss valid-looking substitution, and output monitors miss changes that do not cross a decision boundary.

5.4 Fail-Closed Composition

All six prototype envelopes verify and all eight acceptance checks pass. Table 3 shows that response depends on the coverage contract, not on a single anomaly score. Clean and explicitly approved transpilation are allowed; the latter's semantic-kernel difference is only 7.11×10^{-15}. Parameterized mutation and PSD-preserving kernel substitution are blocked. Prior-preserving label corruption is blocked at the evaluation/report contract despite exactly zero class-prior delta. The approved 256-shot emulator is held for uncertainty adjudication rather than silently allowed.

The prototype ships with a console entry point, tests, CI, a strict output manifest, and reviewer-facing summaries. It remains a local process: there is no network authentication, signed provider record, persistence layer, incident-response evaluation, scheduler binding, or QPU identity.

5.5 Calibrated Coverage and Sensitivity Gates

Three gates, preregistered after the evidence above was frozen (supplement), address two questions the frozen design could not answer: whether nonzero sensor response is detection power, and whether the secondary profile depends on branch-specific scalers or untuned baselines. For each environment, dimension, and model, thresholds for the ten distributional sensors were set at the 191st of 200 clean calibration draws ($\alpha = 0.05$) taken from one half of the held-out evaluation pool and were evaluated on 200 draws from the disjoint other half. Over the 12,000 pooled evaluation draws, per-sensor false-alarm rates lie between 0.029 and 0.069; regime unions reach 0.125, 0.203, 0.048, and 0.259 for $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{Y_m}$, and $\mathcal{I}_{XFY}$ because they join three, six, two, and ten sensors without correction.

Figure 4(a) applies the frozen thresholds to the frozen interventions. The exact blind regions are not threshold artifacts: on all 3,600 evaluation-label observations the calibrated $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ regimes, and on all 1,800 prior-preserving observations the calibrated $\mathcal{I}_{Y_m}$ regime, fire zero

TABLE 3
Executable contract outcomes.

Scenario	Decision	Decisive evidence
Clean	`allow`	All contracts pass
Approved transpilation	`allow`	Policy + semantic equivalence
Parameterized circuit mutation	`block`	Parameter/semantic mismatch
PSD-preserving kernel substitution	`block`	Provenance + semantic mismatch
Prior-preserving label corruption	`block`	Item-aligned outcome evidence
Approved 256-shot emulator	`hold`	Repeat uncertainty review

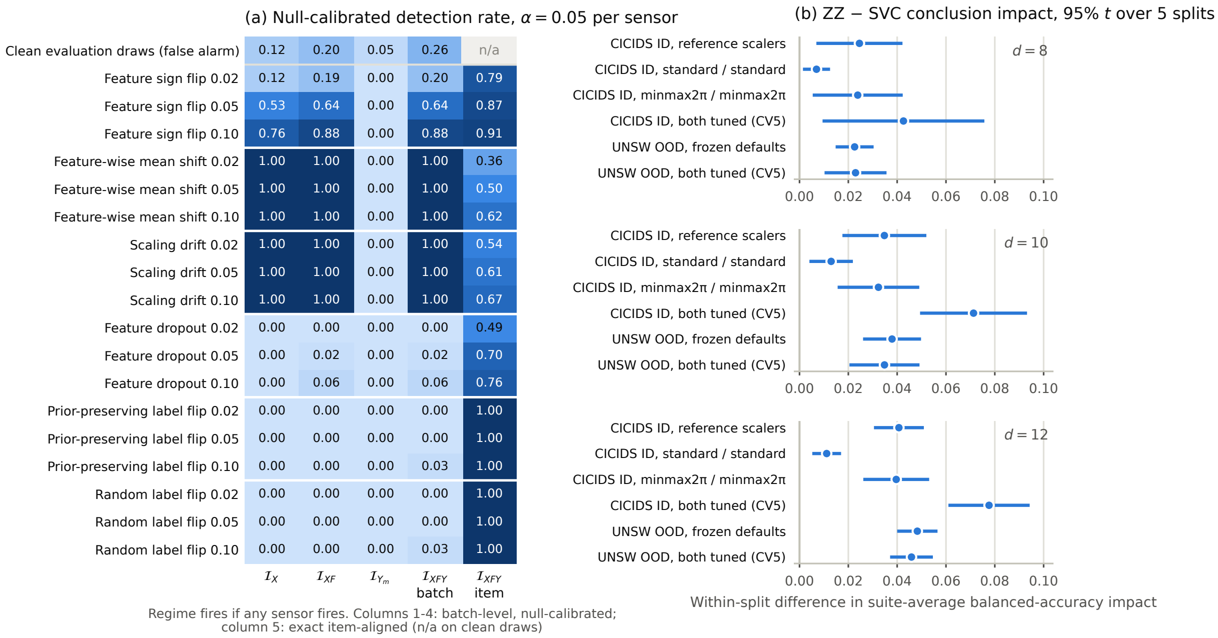


Fig. 4. Preregistered reinforcement gates. (a) Null-calibrated detection rate of each information regime on the frozen interventions, with the empirical false-alarm rate of disjoint clean evaluation draws in the first row; the first four columns are batch-level unions of calibrated distributional sensors, the last column is exact item-aligned detection (undefined on clean draws). (b) Within-split ZZ-minus-SVC conclusion-impact difference with 95% t intervals over five split clusters under three scaler configurations and under cross-validated tuning of both learners.

times, whereas an item-aligned auditor detects every one of the 2,184 positive impacts. A batch-level $\mathcal{I}_{XFY}$ auditor, who sees only distributional confusion statistics without item correspondence, detects 1.8% of them: a 2–10% label corruption lies inside the natural variability of a fresh 128-row batch. Feature-side coverage is mechanism-dependent. Mean shift and scaling drift are detected in every cell at every strength, and so are the two in-place near-null controls (71% and 88%), because the standardized projected features contain tight clusters (in every environment some feature holds 69–87% of rows within 0.01 standard deviations): a fresh batch reproduces the cluster, an in-place perturbation smears it, and the KS sensor reacts. This is a legitimate fingerprint of unmodified data, but a fragile one. Feature dropout with median imputation is nearly invisible to batch-level sensors (0–6%) yet changes predictions in 49–76% of cells.

Figure 4(b) shows that the secondary profile is direction-stable and magnitude-conditional. With matched scalers the within-split ZZ-minus-SVC difference stays positive in 5/5 clusters at every dimension, but standardizing the quantum branch shrinks it from 0.025–0.041 to 0.007–0.013, while matched min–max scaling leaves it at 0.024–0.040. Tuning both learners by five-fold cross-validation on training rows widens it in CICIDS ID (0.043–0.078), tracking the higher clean balanced accuracy that tuning gives ZZ (0.750–0.802 versus 0.741–0.780 for SVC), and leaves UNSW temporal OOD unchanged (0.023–0.046 versus 0.023–0.048). Every configuration reproduces the exact label-path invariances.

6 Integrity-Audit Contract

The evidence from RQ1–RQ3 yields a reporting and enforcement contract. Every integrity claim should name:

1) the workflow boundary and protected asset;
2) the adversary/failure capability, including what remains fixed;
3) the information regime available to the auditor;
4) the sensor, reference, threshold, and false-alarm budget;
5) the covered interventions and exact or empirical blind region; and
6) the response: provenance failure, abstention, rerun, containment, rollback, or human adjudication.

The seven-field implementation record separates boundary and asset into distinct fields and is included in the supplement.

For label-path integrity, the minimum defensive bundle is item-level label provenance plus a joint label-prediction audit. For the studied quantum-kernel boundary it is authenticated circuit/parameter provenance, an approved- transpilation policy, semantic probe kernels, symmetry/diagonal/PSD checks, repeated-estimation evidence, and fail-closed handling of missing or inconsistent evidence. PSD repair may contain a numerical failure but cannot certify integrity. These controls identify the evidence required inside the declared boundary; they do not certify provider or hardware behavior.

The executable decision rule is intentionally asymmetric. `allow` requires every mandatory contract to pass. Expected but unresolved stochastic deviation yields `hold`; a violated invariant or unapproved semantic change yields `block`. Missing mandatory evidence cannot be converted into assurance. This is the operational consequence of information-set conditionality: the system abstains when the preconditions of its integrity claim are absent.

7 Discussion and Limitations

7.1 Security Interpretation

The label-path result exposes a category error with practical consequences. A metric can fall while predictor output is bit-for-bit unchanged. Calling that model degradation assigns failure to the wrong component and suggests the wrong defense: input robustness cannot repair corrupted ground truth; provenance and joint validation can.

Likewise, zero sensor response is informative only if the sensor receives evidence capable of distinguishing the states. A circuit hash cannot expose a calibration change, a PSD check cannot establish label provenance, and an output monitor cannot protect circuit confidentiality. Hybrid assurance therefore requires a coverage argument across boundaries, not one global anomaly score.

The secondary ZZ profile illustrates why impact must remain separate from auditability. Its direction is consistent in most fixed environments, but magnitude and interval support vary with clean headroom, scaling, and feature map. Causal attribution to quantum geometry would require symmetric preprocessing, headroom-aware endpoints, or kernel-level decomposition. No such universal claim is made here.

7.2 Validity Boundaries

The study has eight principal limitations. First, three datasets and eight scale/temporal settings improve transport evidence but remain fixed cybersecurity environments, not sampled deployments. Five split clusters measure within-environment sampling only. Second, most gates use 128/128 samples; the 256/256 gate is a scale sensitivity, not large-scale QML evidence. Third, balanced numeric-only staging does not reproduce operational prevalence or categorical-feature handling.

Fourth, both quantum evaluators are ideal statevector simulators. The separate gate adds binomial shot emulation, not a sampler backend, device-calibrated noise, or hardware execution. Fifth, SVC and QSVC retain pipeline-specific scalers, so their comparison does not isolate kernel geometry; the matched- scaler and tuned gates show that the direction persists while the magnitude moves with the quantum-branch scaler and with clean headroom. Sixth, exact blindness is a capability result under a constructed perturbation; it neither estimates deployment frequency nor implies malicious intent.

Seventh, the calibrated rates are fixed-design, simulator-only sensitivities: thresholds are pooled over the splits of one staged table, regime unions are uncorrected, the largest environment reuses a small pool, and the KS sensor's sensitivity rests on cluster structure that an adversary who preserves it would evade. Min–max stealth remains a descriptive sensitivity. Deployment still requires context-specific null models, false-alarm budgets, and power analysis. Eighth, the prototype is not a deployed service or authenticated provider ledger. Its hash chain is tamper-evident but supplies neither identity nor non-repudiation. Scheduling, provider security, multi-tenancy, confidentiality, physical QPU execution, and operational Fleet Management are excluded rather than silently treated as future evidence for this study.

8 Conclusion

Hybrid workflow integrity cannot be reduced to a robustness number. An intervention can change a reported conclusion without changing the model, and a sensor can remain silent because its evidence is mathematically invariant. We formalized this dependency as information-set conditional auditability, derived exact blind regions for label-only and prior-preserving interventions, and reproduced those boundaries across CICIDS2017, UNSW-NB15, ToN-IoT, and eight fixed expansion environments. A bounded quantum simulator gate then showed why provenance, semantic, algebraic, repeated-estimation, and output controls are complementary; a local contract converted that coverage into fail-closed decisions. Preregistered calibration confirmed that the blind regions are not threshold artifacts and that the secondary model profile is pipeline- conditional.

The resulting requirement is precise: every integrity claim must state its boundary, information set, sensor coverage, residual blind region, calibration, and response. Assurance is granted only when the evidence required by that claim is present and consistent. The evidence is complete for this declared research boundary and deliberately supports no full-lifecycle, provider, hardware, or Fleet Management security claim.

Data, Code, and Ethics Statement

The review artifact contains source code, a locked environment, tests, compact derived tables, figures, SHA-256 manifests, and exact reproduction commands. Raw CICIDS2017, UNSW-NB15, and ToN-IoT data are not redistributed; their public sources and deterministic staging procedures are documented. The artifact DOI will be inserted after author, license, and repository-visibility confirmation. The study uses previously released network-security benchmarks and involves no new human-participant, animal, or personal-data collection.

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